

Nompumelelo Nkikande

BrightLearn SQL Aggregate Functions & Operators
Exercise 02

Q1: SELECT DISTINCT department
FROM Students;

department
IT
HR
Finance

Q2: SELECT department,
AVG(age) AS avg_age
FROM Students
GROUP BY department;

department	avg_age
IT	20.5
HR	22
Finance	23

Q3: SELECT 'department',
COUNT(Student_id) AS Student_count
FROM Students
GROUP BY department
HAVING COUNT(Student_id) > 1;

department	Student_count
IT	2
HR	2

Q4: **SELECT** ~~Student_id~~ *
FROM Students
WHERE age **BETWEEN** 21 **AND** 23;

Student_id	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5: **SELECT** *
FROM Students
WHERE department **IN** ('IT', 'HR')
AND age > 21;

Student_id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q6: **SELECT** department,
SUM(credits) **AS** total_credit
FROM courses
GROUP BY department
HAVING **SUM**(credits) > 5;

department	total_credits
IT	11

Q7: **SELECT** *
FROM courses
WHERE credits < 4;

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
103	Statistics	HR	3

Q8: **SELECT** Course_id, Course_name, ~~Course~~ credits
FROM courses
ORDER BY credits **DESC**
LIMIT 3;

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Q9: **SELECT** grade,
MAX(grade) AS Max_grade
MIN(grade) AS Min_grade
AVG(grade) AS avg_grade
FROM enrollments;

Max_grade	Min_grade	avg_grade
90	78	84.6

Q10: **SELECT** ^{course_id,}
COUNT(^{enrollment} ~~course_id~~) **AS** enrollment_count
FROM courses; X question says PER course
GROUP BY course_id, →

Course_id	Enrollment_count
101	1
102	1
103	1
104	1
105	1

Q11: **SELECT** department
SUM(Salary) **AS** total_salary
SUM(bonus) **AS** total_bonus
FROM Salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

Q12: **SELECT** department,
AVG(salary) **AS** avg_salary
FROM Salaries
GROUP BY department
HAVING **AVG**(salary) > 55000;

department	avg_salary
IT	61000
Finance	70000

Q13. **SELECT** employee_id, name, salary, bonus
SUM (salary **AND** bonus) **AS** total_compensation
FROM Salaries
~~**WHERE** salary > 60000;~~ **AND** *GROUP BY? salary
HAVING SUM (Salary **AND** bonus) > 60000

employee_id	name	Salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Q14: **SELECT** department,
SUM (budget), **AS** total_budget,
AVG (budget) **AS** total_avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;

department	total_budget	avg_budget
IT	270000	135000

Q15: **SELECT** *
FROM projects
~~**WHERE** budget **BETWEEN** 50000 **AND** 120000~~
WHERE NOT Marketing
AND budget **BETWEEN** 50000 **AND** 120000;

Project_id	project_name	department	budget
1	AI app	IT	120000
2	Payroll System	Finance	80000
3	HR Portal	HR	50000